MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSCCSC202 - Python for Data Analytics

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each Course Outcome (CO) to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

(Answer all questions, each question carries 2 marks.)

1. List the main steps in data analysis process. [R] [CO1]
2. Compare structured and unstructured data. [U] [CO1]
3. Let arr be a NumPy array. Explain the result of the operation arr * arr. [U] [CO2]
4. Differentiate between isna() and notna() functions in Pandas. [U] [CO3]
5. Apply head() function to display the first two rows of a DataFrame created with data of your choice. [A] [CO3]

[5 x 2 = 10]

Part B

Short Answer Questions

*(Answer **5** out of 7 questions, each question carries **5** marks.)*

6. Differentiate predictive and prescriptive data analysis with a suitable example. [U] [CO1]
7. Explain how slicing works in 1D arrays with code examples. [U] [CO2]
8. Discuss the working of empty() and full() in numpy with examples. [U] [CO2]
9. Explain how Boolean indexing can be used to filter positive numbers from an array containing both positive and negative values. Provide code and output. [U] [CO2]
10. Differentiate between Series and DataFrame in terms of data handling. [U] [CO3]
11. Write a Python program to compute mean, median, and standard deviation of a Pandas Series. [A] [CO3]
12. Given a DataFrame df with columns ["Student","Marks","Grade"], write code to select rows with labels 1 and 3 and only the columns "Student" and "Marks". Show output. [A] [CO3]

[5 x 5 = 25]

Part C

Essay Questions

*(Answer **1** out of 2 questions, each question carries **15** marks.)*

13. Discuss the concept of ndarray in NumPy. Explain in detail the different methods of creating ndarrays along with their syntax and suitable examples. [U] [CO2]

14. Describe how Matplotlib can be used for data visualization. Explain the purpose and usage of any five plot types with examples. [U] [CO3]

[1 x 15 = 15]